SkyBridge Alaska

Aviation Safety Mesh Network - Saving Lives Through Community-Powered Infrastructure

Learn More

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The Aviation Crisis in Alaska

Alaska's vast and rugged terrain, coupled with extreme weather, creates perilous flying conditions. Pilots often operate in remote areas without reliable radio communication, making them vulnerable to unexpected hazards and unable to call for help during emergencies. This communication gap is a leading cause of aviation incidents and fatalities.

When traditional radio systems fail, pilots are left isolated. SkyBridge Alaska addresses this critical vulnerability by establishing a robust, community-powered mesh network, ensuring continuous communication and greatly enhancing safety for everyone in the air.

The Aviation Crisis in Alaska

36x

Higher Death Rate

Alaska pilots are 36 times more likely to die than the average US worker according to the CDC

263

Mapping Errors

Terrain mapping errors up to 263 feet contribute to fatal crashes

15

Fatal Crashes

Controlled flight into terrain crashes since 2008, killing 16 people

As The Washington Post investigation revealed, terrain mapping errors directly contribute to fatal crashes. One expert noted that "Mars is better mapped than the state of Alaska." Traditional government solutions remain stuck in budget gridlock after decades, with $30 million needed just to finish mapping Alaska.

The $50 Solution That Changes Everything

Simple Technology

SkyBridge uses affordable $50 Meshtastic radios that plug into aircraft power. No expensive avionics, no monthly fees, no corporate overlords bleeding you dry.

Your phone becomes your complete aviation services portal with the SkyBridge app running on iPhone or Android devices you already carry.

Community Network

Every pilot who joins makes coverage better for everyone. More reliability, more safety, more information. You're not just buying a radio, you're joining a community that has each other's backs.

Share your location if you want, keep it private if you don't. The network works for you, not against you.

Never Miss Critical Radio Calls Again

01

Alaska DOT&PF Base Stations

Capture every VHF transmission across Alaska's airspace

02

Mesh Network Distribution

Critical information sent instantly through the mesh network

03

Text to Your Phone

Receive weather reports and traffic advisories as text messages

You know that sinking feeling when you realize you missed THE critical weather report or traffic advisory that was said once when you were out of range. With SkyBridge, you'll always know what was said, even when flying in remote areas covering 80% of Alaska where traditional infrastructure fails.

Proven Technology Stack

Hardware

Affordable LoRa radios available from multiple vendors, specifically designed for situations where everything else fails

Protocol

NASA TAIGA ASN.1 for efficient data compression and Meshtastic open-source mesh networking

Interface

Mobile app for iOS and Android - no cell towers or satellites required

Don't let the name fool you. Meshtastic is serious technology used by emergency responders, hikers, and professionals worldwide. It's proven, reliable, and specifically designed for critical communications when traditional systems fail.

Project Status: Operational Prototypes

1. **Working Prototypes**

   Operational Meshtastic devices deployed and tested with pilots successfully exchanging messages

2. **Alaska DOT&PF Partnership**

   Active pilot program scaling statewide with government support

3. **Patent Protection**

   Three provisional patents filed protecting core innovations, enabling confident commercial partnerships

4. **Industry Collaboration**

   Active partnerships with Meshtastic and Rokland Technologies for advanced integration

Advanced Integration Capabilities

All-in-One Radio Solution

Rokland Technologies partnership developing VHF/ADS-B/SDR/LoRa combination units for seamless aircraft integration

Aircraft Systems Integration

CANBUS/OBD2/ARINC 429 connectivity enables automated reporting and real-time aircraft status updates

VHF Transcription

Ground stations automatically convert radio chatter to text and distribute via Meshtastic mesh network

Multi-Agency Support

RWIS highway weather, UAF Volcanic Institute, USGS seismic networks all supported through unified platform

Multi-State Expansion Ready

Rural Aviation Challenges

Montana, Idaho, Wyoming, Colorado face similar terrain and weather risks as Alaska

Economic Viability

$50 nodes vs $200K ground stations - economically viable for rural airports and pilot communities

State Control

Community-operated infrastructure under state oversight, not federal dependency

Interstate Cooperation

Shared development costs, shared safety benefits across participating states

Ready for coordinated deployment nationwide with working Meshtastic prototypes, NASA TAIGA protocol integration proven, and Alaska DOT&PF partnership established. Seeking 3-5 additional states for multi-state pilot program.

Dual Licensing Model

Free Use (AGPL-3.0)

- State and federal agencies (Alaska DOT&PF, FAA, NOAA, NWS)
- Search and rescue organizations
- Educational institutions
- Small Part 135 operators (fewer than 5 aircraft)
- 501(c)(3) nonprofits serving Alaska aviation

Commercial License Required

- Part 135 operators with 5+ aircraft
- Part 121 air carriers
- Oil & gas, mining, tourism companies
- Out-of-state commercial operators
- Technology companies creating derivative products

This dual licensing approach supports both public safety and commercial innovation, ensuring the core system remains open source while generating revenue for ongoing development through commercial partnerships.

Contact & Resources

Project Leadership

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Key Resources

- Technical Architecture & System Specifications
- NASA TAIGA Protocol Documentation
- Hardware Specifications & Component Costs
- Alaska Aviation Gap Analysis
- Complete Technical Whitepaper

Project Links

https://skybridgealaska.net

GitHub Repository

Built by Alaska DOT&PF using Meshtastic technology - proven, reliable, and specifically designed for situations where everything else fails.